

Electrical Machine- II (BT EE403A)

UNIT-III

Syllabus/Topics Covered

DFIG induction machine, control strategies for a DFIG: Vector control, Magnitude and frequency control. Synchronized model of grid connected doubly fed induction generator for wind power generation, pitch control analysis, mechanical characteristics analysis and torque-slip characteristics by Matlab.

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4. DOUBLY FED INDUCTION GENERATOR

Currently DFIG wind turbines are increasingly used in large wind farms. A typical DFIG system is shown in the below figure. The AC/DC/AC converter consists of two components: the rotor side converter C_{rotor} and Grid side converter C_{grid} . These converters are voltage source converters that use forced commutation power electronic devices (IGBTs) to synthesize AC voltage from DC voltage source. A capacitor connected on DC side acts as a DC voltage source. The generator slip rings are connected to the rotor side converter, which shares a DC link with the grid side converter in a so called back-to-back configuration. The wind power captured by the turbine is converted into electric power by the IG and is transferred to grid by stator and rotor windings. The control system gives the pitch angle command and the voltage commands for C_{rotor} and C_{grid} to control the power of the wind turbine, DC bus voltage and reactive power or voltage at grid terminals.

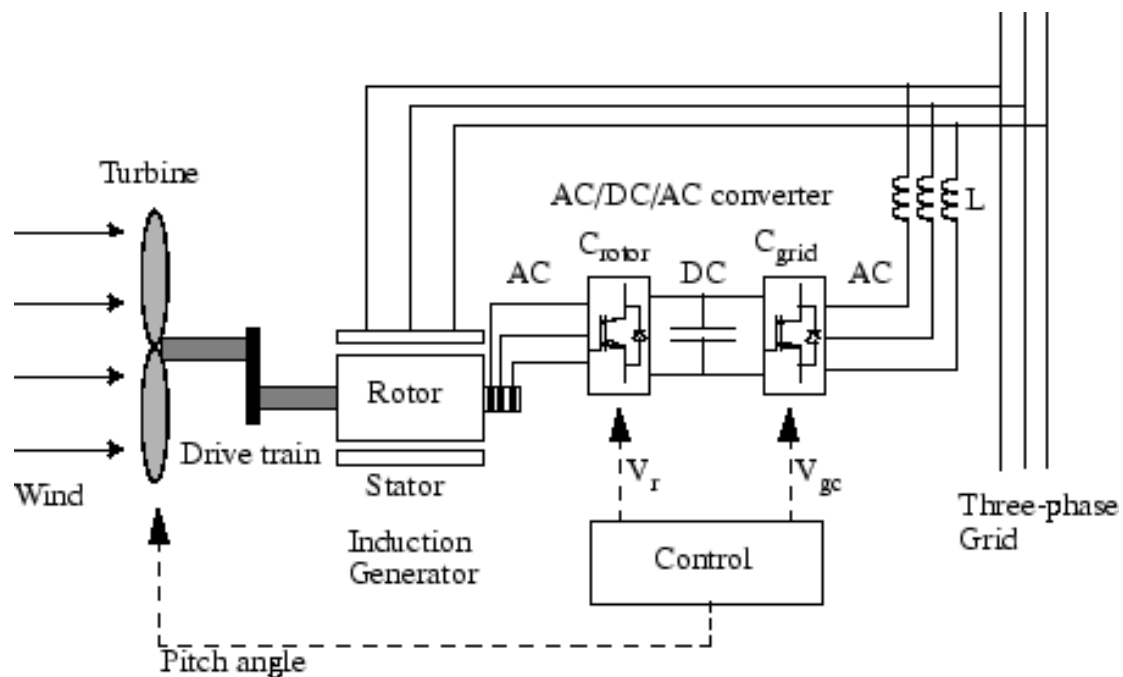


Fig4.1 : A DFIG and wind turbine system

OPERATION:

When the rotor speed is greater than the rotating magnetic field from stator, the stator induces a strong current in the rotor. The faster the rotor rotates, the more power will be transferred as an electromagnetic force to the stator, and in turn converted to electricity which is fed to the electric grid. The speed of asynchronous generator will vary with the rotational force applied to it. Its difference from synchronous speed in percent is called generator's slip. With rotor winding short circuited, the generator at full load is only a few percent.

With the DFIG, slip control is provided by the rotor and grid side converters. At high rotor speeds, the slip power is recovered and delivered to the grid, resulting in high overall system efficiency. If the rotor speed range is limited, the ratings of the frequency converters will be small compared with the generator rating, which helps in reducing converter losses and the system cost.

Since the mechanical torque applied to the rotor is positive for power generation and since the rotational speed of the magnetic flux in the air gap of the generator is positive and constant for a constant frequency grid voltage, the sign of the rotor electric power output is a function of the slip sign. C_{rotor} and C_{grid} have the capability of generating or absorbing reactive power and can be used for controlling the reactive power or the grid terminal voltage. The pitch angle is controlled to limit the generator output power to its normal value for high wind speeds. The grid provides the necessary reactive power to the generator.

4.1 Steady state characteristics:

The steady state performance can be explained using Steinmetz per phase equivalent circuit model as shown in figure where motor convention is used. In this figure v_s and v_r are the stator and rotor voltages, i_s and i_r are the stator and rotor currents, r_s and r_r are the stator and rotor resistances (per phase), X_s and X_r are stator and rotor leakage reactance's, X_m is the magnetizing reactance and s is slip.

The steady state equivalent circuit of DFIG is shown in Fig.4.2

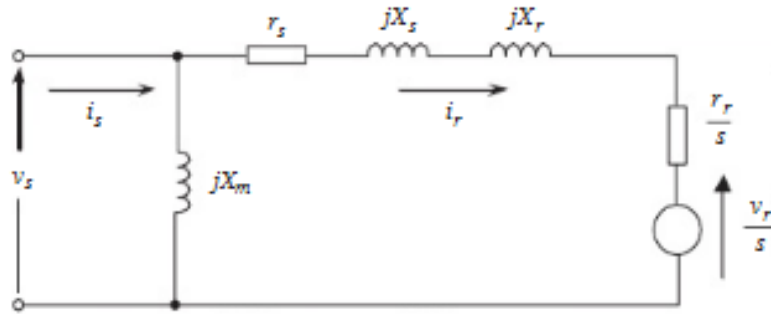


Fig: 4.2 steady state equivalent circuit of DFIG

To obtain the torque equation from the equivalent circuit, we can simplify the steady state induction motor circuit by moving X_m to the stator terminal. The rotor current I_r is expressed as

$$I_r = \frac{V_s - \left(\frac{V_r}{s}\right)}{\left(r_s + \frac{r_r}{s}\right) + j(X_s + X_r)}$$

The electrical torque T_e , from the power balance across the stator to rotor gap, can be calculated from

$$T_e = \left(I_r^2 \frac{r_r}{s}\right) + \frac{P_r}{s}$$

Where the power supplied or absorbed by the controllable source injecting voltage into the rotor circuit, that is the rotor active power, P_r can be calculated from

$$P_r = \frac{V_r}{s} I_r \cos \theta$$

$$P_r = \operatorname{Re} \left(\frac{V_r}{s} I_r^* \right)$$

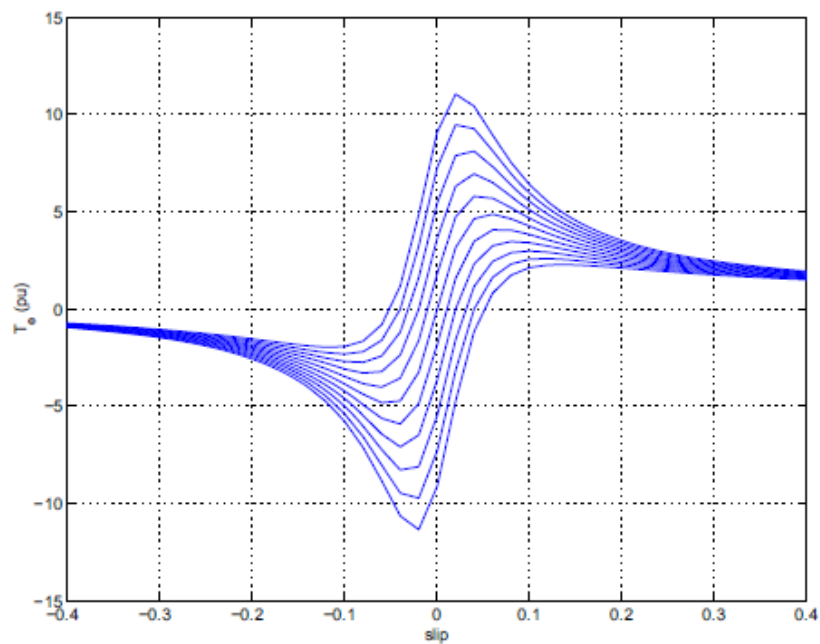
TORQUE-SLIP CHARACTERISTICS OF DFIG:

Fig: 4.3 Torque-slip characteristic when the angle of V_r is 0.
 $|V_r|$ is changing from -0.05 to +0.05 pu.

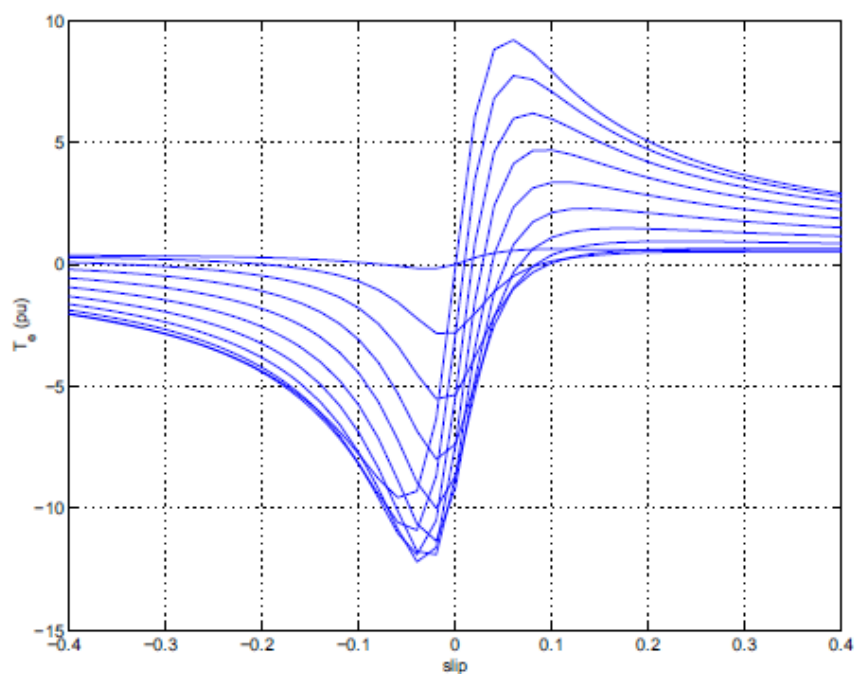


Fig: 4.4 Torque-slip characteristic when $|V_r|$ is 0.05 pu.
 The angle of V_r is changing from -90° to $+90^\circ$.

4.2 CONTROL STRATEGIES FOR A DFIG:

1. Vector control
2. Magnitude and frequency control

4.2.1 VECTOR OR FIELD ORIENTED CONTROL THEORY:

The complete control strategy of the machine is divided in two ways, one is scalar control and the other is vector control. The limitations of scalar control give a significance to vector control. Though the scalar control strategy is modest to implement but the natural coupling effect gives sluggish response. The inherent problem is being solved by the vector control. The vector control is invented in the beginning of 1970s. Using this control strategy an IM can be performed like dc machine. Because of dc machine like performance vector control is also known as orthogonal, decoupling or Tran's vector control. Different Vector control strategies have been proposed to control the active and reactive power of an induction generator.

The basic of the vector control theory is d-q theory. To understand vector control theory knowledge about d-q theory is essential.

D-Q THEORY:

The d-q theory is also known as reference frame theory. The history says in 1920, R. H. Park suggested a new theory to overcome the problem of time varying parameters with the ac machines. He formulated a change of variables which replace the variables related to the stator windings of a synchronous machine with variables related with fictitious winding which rotates with the rotor at synchronous speed.

Essentially he transformed the stator variables to a synchronously rotating reference frame fixed in the rotor. With such transformation (Park's transformation) he showed that all the time varying inductances that occur due to an electric circuit in relative motion and electric circuit with varying magnetic reluctances can be eliminated. Later in 1930s H. C. Stanley showed that time varying parameters can be eliminated by transforming the rotor variables to the variables associated with fictitious stationary windings.

In this case the rotor variables are transformed to the stationary reference frame fixed on the stator. Later G. Kron proposed transformation of stator and rotor variables to a synchronously rotating reference frame which moves with rotating magnetic field. Latter, Krause and Thomas had shown that the time varying Inductances can be eliminated by referring the stator and rotor variables to an arbitrary reference frame which may rotate at any speed [5].

TRANSFORMATION OF THREE PHASE STATIONARY TO TWO PHASE STATIONARY AXES:

Consider a symmetrical three phase induction machine with stationary a-phase, b phase and c-phase axes are placed at 120° angle to each other as shown in Fig 4.5. The main aim is to transform the three phase stationary frame variables into two phase stationary frame variables (d^s - q^s) and then transform these to synchronously rotating reference frame variables (d-q), and vice versa.

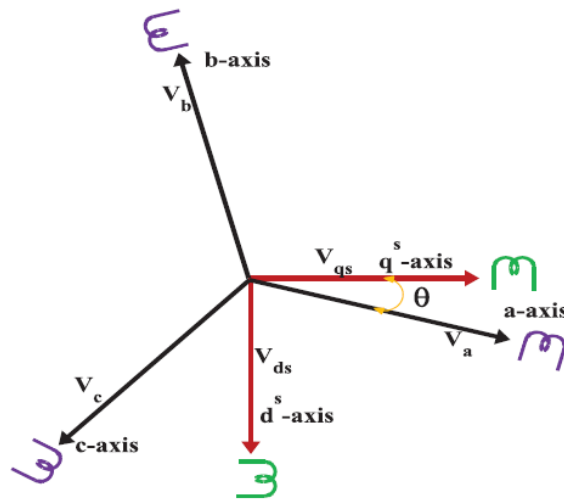


Fig 4.5: Transformation of a-b-c to d^s - q^s axes

Let d^s - q^s axes are oriented at an angle θ from a-b-c axes as shown in Fig 4.2 The voltage (V_{ds}^s and V_{qs}^s) can be resolved into a-b-c components and can be represented in the matrix form as

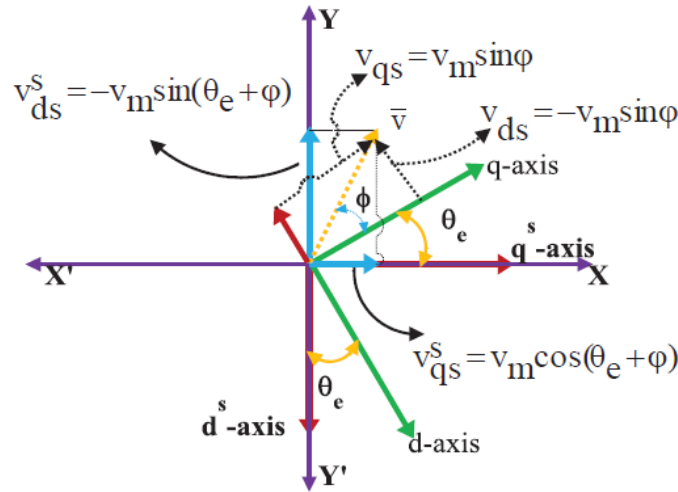
$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 1 \\ \cos(\theta - 120^\circ) & \sin(\theta - 120^\circ) & 1 \\ \cos(\theta + 120^\circ) & \sin(\theta + 120^\circ) & 1 \end{bmatrix} \begin{bmatrix} V_{qs}^s \\ V_{ds}^s \\ V_{0s}^s \end{bmatrix} \quad (4.1)$$

The corresponding inverse relation is

$$\begin{bmatrix} v_{qs}^s \\ v_{ds}^s \\ v_{0s}^s \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin \theta & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (4.2)$$

Where v_{0s}^s is added as the zero sequence component. Other parameters like current, flux linkages can be transformed by similar manner. It is more convenient to set $\theta=0^\circ$, so that q-axis is aligned with the a-axis in this case (The alignment of the axes are optional, d-axis can also be aligned with a-axis). The sine components of d and q parameters will be replaced with cosine values, and vice versa if d-axis coincides with a-axis.

Transformation of two phase stationary axes to two phase synchronously rotating axes:



4.6 Transformation of stationary d^s - q^s axes to synchronously rotating frame d-q axes

Fig 4.6 above shows the synchronously rotating d-q axes which rotate at synchronous speed ω_e with respect to d^s - q^s axes. The two phase windings are transformed in to the fictitious windings mounted on the d-q axes.

The voltages on the d^s - q^s axes can be converted into d-q axes as follows;

$$v_{qs} = v_{qs}^s \cos \theta_e - v_{ds}^s \sin \theta_e \quad (4.3)$$

$$v_{ds} = v_{qs}^s \sin \theta_e + v_{ds}^s \cos \theta_e \quad (4.4)$$

Again resolving the rotating frame parameters into a stationary frame the relations are

$$V_{qs}^s = v_{qs} \cos \theta_e + v_{ds} \sin \theta_e \quad (4.5)$$

$$V_{ds}^s = -v_{qs} \sin \theta_e + v_{ds} \cos \theta_e \quad (4.6)$$

Mathematical modelling of Induction Generator:

In this section the basic mathematical modelling of DFIG is described in detail. From the previous section we confirm that the three phase parameters can be represented in two phase parameters and vice versa using certain fundamental rules. In this section the machine modelling is explained by taking two phase parameters into consideration. Though the basic concepts behind the DFIG system is explained briefly in short we can say the DFIG is a wound rotor type induction machine, its stator consists of stator frame, stator core, poly phase (3-phase) distributed winding, two end covers, bearing etc. The stator core is made up of stack of cylindrical steel laminations which are slotted along their inner periphery for covering the 3-phase winding. Its rotor consists of slots in the outer periphery to house the windings like stator. The machine works on the principle of Electromagnetic Induction and the energy transfer takes place by means of transfer action. So the machine can represent as a transformer but rotatory not stationary.

Modelling of DFIG in synchronously rotating frame:

The equivalent circuit diagram of an induction machine is shown in Fig.4.7 and Fig.4.8. In this figure the machine is represented as two phase machine, it has already been discussed before that a three phase machine can be represented as two phase machine obeying certain rules. For the modelling of DFIG in synchronously rotating frame we need to represent the two phase stator (d^s - q^s) and rotor (d^r - q^r) circuit variables in a synchronously rotating (d - q) frame.

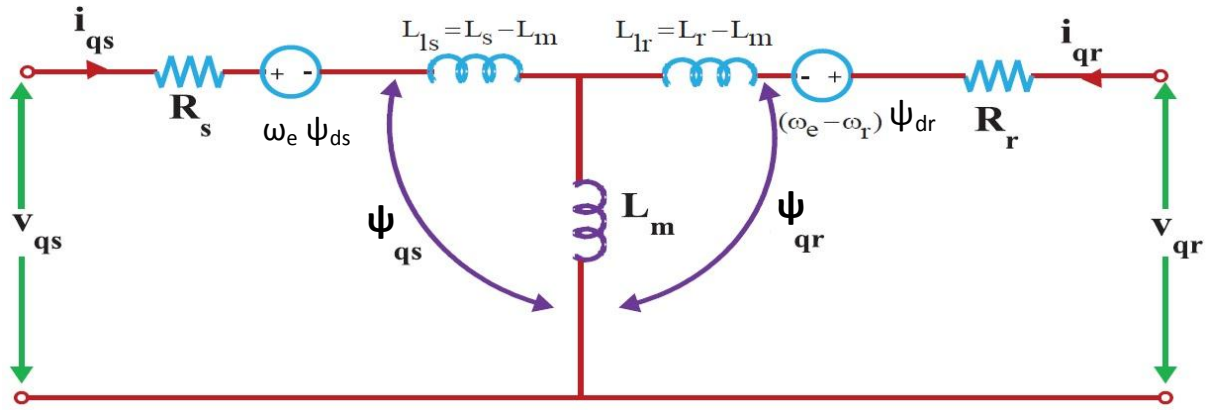


Fig.4.7 Dynamic d-q equivalent circuit of DFIG (q-axis circuit)

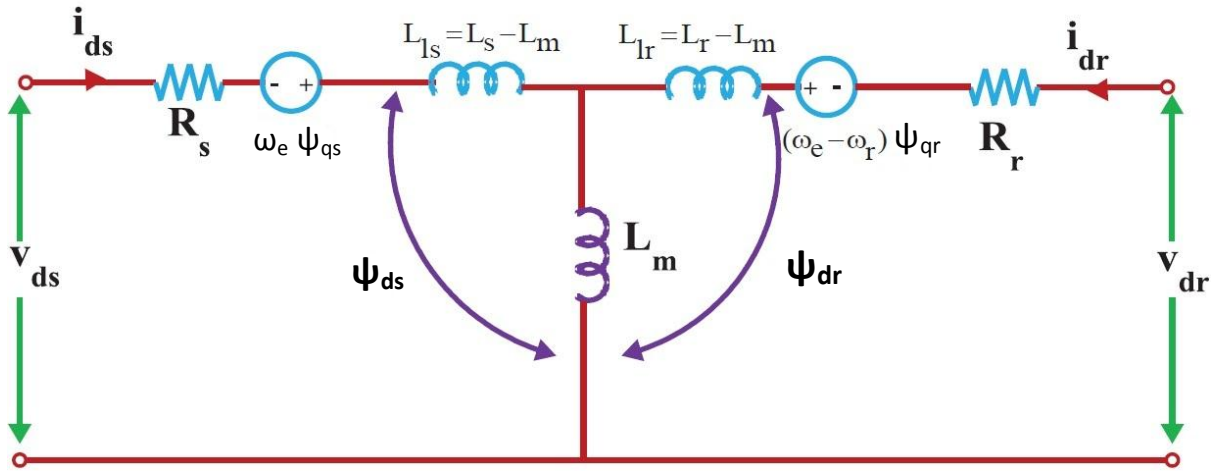


Fig.4.8 Dynamic d-q equivalent circuit of DFIG (d-axis circuit)

The stator circuit equations are given below:

$$v_{qs}^s = R_s i_{qs}^s + \frac{d}{dt} \psi_{qs}^s \quad (4.7)$$

$$v_{ds}^s = R_s i_{ds}^s + \frac{d}{dt} \psi_{ds}^s \quad (4.8)$$

Where ψ_{qs}^s and ψ_{ds}^s are q-axis and d-axis stator flux linkages, respectively.

Converting Eq. (4.7) and Eq. (4.8) to d-q frame the following equations can be written as:

$$v_{qs} = R_s i_{qs} + \frac{d}{dt} \psi_{qs} + \omega_e \psi_{ds} \quad (4.9)$$

$$v_{ds} = R_s i_{ds} + \frac{d}{dt} \psi_{ds} - \omega_e \psi_{qs} \quad (4.10)$$

Where all the variables are in synchronously rotating frame. The bracketed terms are defined as the back e.m.f. or speed e.m.f or counter e.m.f. due to the rotation of axes as in the case of

DC machines. When the angular speed ω_e is zero the speed e.m.f due to d and q axis is zero and the equations changes to stationary form.

Owing to the rotor circuit, if the rotor is blocked or not moving, i.e. $\omega_r=0$, the machine equations can be written in similar way as stator equations:

$$v_{qr} = R_r i_{qr} + \frac{d}{dt} \psi_{qr} + \omega_e \psi_{dr} \quad (4.11)$$

$$v_{dr} = R_r i_{dr} + \frac{d}{dt} \psi_{dr} - \omega_e \psi_{qr} \quad (4.12)$$

All the parameters are referred to the primary circuit, which is a stator in this case. Let the rotor rotates at an angular speed ω_r , then the d-q axes fixed on the rotor fictitiously will move at a relative speed $\omega_e - \omega_r$ to the synchronously rotating frame.

The d-q frame rotor equations can be written by replacing $\omega_e - \omega_r$ in place ω_e of as follows:

$$v_{qr} = R_r i_{qr} + \frac{d}{dt} \psi_{qr} + (\omega_e - \omega_r) \psi_{dr} \quad (4.13)$$

$$v_{dr} = R_r i_{dr} + \frac{d}{dt} \psi_{dr} - (\omega_e - \omega_r) \psi_{qr} \quad (4.14)$$

The flux linkage expressions in terms of current can be written from Fig.4.7 and Fig.4.8 as follows:

$$\psi_{qs} = L_{ls} i_{qs} + L_m (i_{qs} + i_{qr}) = L_s i_{qs} + L_m i_{qr} \quad (4.15)$$

$$\psi_{ds} = L_{ls} i_{ds} + L_m (i_{ds} + i_{dr}) = L_s i_{ds} + L_m i_{dr} \quad (4.16)$$

$$\psi_{qr} = L_{lr} i_{qr} + L_m (i_{qs} + i_{qr}) = L_r i_{qr} + L_m i_{qs} \quad (4.17)$$

$$\psi_{dr} = L_{lr} i_{dr} + L_m (i_{ds} + i_{dr}) = L_r i_{dr} + L_m i_{ds} \quad (4.18)$$

$$\psi_{qm} = L_m (i_{qs} + i_{qr}) \quad (4.19)$$

$$\psi_{dm} = L_m (i_{ds} + i_{dr}) \quad (4.20)$$

Eq. (4.7) to Eq. (4.20) describes the complete electrical modelling of DFIG. Whereas the Eq. (4.21) expresses the relations of mechanical parameters which are essential part of the modelling.

The electrical speed ω_r cannot be treated as constant in the above equations. It can be connected to the torque as

$$T_e = T_L + J \frac{d\omega_m}{dt} + B\omega_m = T_L + \frac{2}{p} J \frac{d\omega_r}{dt} + \frac{2}{p} B\omega_r \quad (4.21)$$

PRINCIPLE OF VECTOR CONTROL:

The fundamentals of implementation of vector control technique can be explained using the Fig 4.9 In this figure the machine model is in synchronously rotating frame. The vector control uses unit vectors to obtain the appropriate control action. The main role of unit vector is to convert the 2-phase model to 3-phase model and vice versa. Though the control techniques used for DFIG uses two axes parameters as explained in the modelling via vector control but the model is virtual representation of the original machine. The control signals which will be fed to the original machine or converters should be in three axes form, so the process requires repeated conversion of two-phase to three-phase parameter or vice versa following the necessary action being taken for the system [6]. There are essentially two general method of vector control

- Direct or feedback method (which is invented by Blaschke)
- Indirect or feed forward method (which is invented by Hasse)

The two methods are different from each other by the process of generating unit vector for control. Unit vectors ($\cos\theta_e$, $\sin\theta_e$) are generally generated using the flux vectors, but it can also be generated using voltage vectors. The name of the orientation of unit vector is given according to the vector taken for generation of θ_e . The names of the orientations used are given below.

- Rotor flux orientation
- Stator flux orientation
- Air gap flux orientation

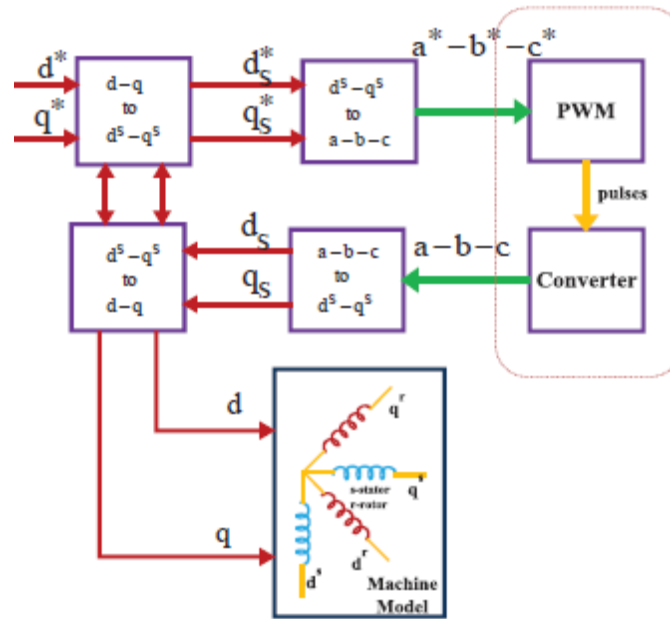


Fig 4.9 Implementation of vector control principle

The detail vector control strategy is shown in above figure. The a, b, c components are generated from the controlled components a^* , b^* , c^* respectively using vector control techniques. The machine terminal parameters (either voltages or currents) are converted to d^s - q^s components by 3-phase to 2-phase transformation. These are then converted to synchronously rotating frame by the unit vector before applying to the 2-phase machine model. The controller makes two stage of inverse transformation as shown, so that the control components d^* and q^* corresponds to the machine parameters d and q respectively.

4.2.2 SYNCHRONISED MODEL OF GRID CONNECTED DOUBLY FED INDUCTION GENERATOR FOR WIND POWER GENERATION:

MAGNITUDE AND FREQUENCY CONTROL OF DFIG:

A magnitude and frequency control (MFC) strategy has been proposed for the grid connected doubly fed induction generator (DFIG). The proposed MFC makes the DFIG equivalent to a synchronous generator in the power system. The active and reactive powers of the stator depend on the phase and magnitude of the new equivalent 'emf behind the internal transient reactance'. The relationship between the rotor voltage and the 'emf behind the internal transient reactance' is also detailed. Unlike traditional control strategies such as stator-flux-orientation vector control and FMAC, the MFC method manipulates the magnitude and

frequency of the rotor voltage. This simplifies the design of the control system and improves system reliability. Thus, co-ordinate transformations, rotor position detection, and measurements of rotor currents and rotor speeds are not required [3].

SYNCHRONISED MODEL OF DFIG:

MODELLING OF DFIG STATOR:

It is assumed that the stator transient can be neglected in this paper. The effects of neglecting stator transients in DFIG model were analysed. Besides analysis, includes simulated waveforms which establish that the stator transients in DFIG can be neglected and the accuracy is not affected after the transients have damped out.

By neglecting the stator transient, the voltage equations of the DFIG in the arbitrary d-q reference frame can be expressed as follows (stator in generator convention and rotor in motor convention)

$$\begin{aligned} u_{d1} &= -r_1 i_{d1} - \psi_{q1} \omega_1 \\ u_{q1} &= -r_1 i_{q1} - \psi_{d1} \omega_1 \\ u_{d2} &= -r_2 i_{d2} + p\psi_{d2} - \psi_{q2} \omega_2 \\ u_{q2} &= -r_2 i_{q2} + p\psi_{q2} - \psi_{d2} \omega_2 \end{aligned}$$

The corresponding flux linkage equations:

$$\begin{aligned} \psi_{d1} &= -L_1 i_{d1} - L_m i_{d2} \\ \psi_{q1} &= -L_1 i_{q1} + L_m i_{q2} \\ \psi_{d2} &= L_2 i_{d2} - L_m i_{d1} \\ \psi_{q2} &= L_2 i_{q2} - L_m i_{q1} \end{aligned}$$

Setting the d-axis to align with the rotor flux vector, one defines $\psi_2 = \psi_{d2}$. A consequence of the rotor flux alignment is $\psi_{q2} = 0$.

Thus, rotor currents can be expressed in terms of stator currents as:

$$\begin{aligned} i_{d2} &= \frac{\psi_2 + L_m i_{d1}}{L_2} \\ i_{q2} &= \frac{L_m}{L_2} i_{q1} \end{aligned}$$

In order to eliminate the rotor variables in stator equations, define

$$E'_q = \omega_1 \frac{L_m}{L_2} \psi_2$$

$$X'_1 = \sigma \omega_1 L_1$$

Where E'_q is the equivalent 'emf behind the internal transient reactance' which is generated by the rotor flux linkage ψ_2 , X'_1 is the transient reactance of the stator, and $\sigma = (L_1 L_2 - L_m^2) / L_1 L_2$ is the leakage factor.

By substituting, the stator voltage equations can be written as follows:

$$u_{d1} = -r_1 i_{d1} + X'_1 i_{q1}$$

$$u_{q1} = -r_1 i_{q1} - X'_1 i_{d1} + E'_q$$

Neglecting the stator resistance, the vector diagram of the DFIG stator can be drawn as shown in Fig 4.7 according to above equations.

In this vector diagram, δ is the power angle between the vector E'_q and U_1 and ϕ is the phase angle between the vector U_1 and I_1 . Based on Fig 4.7, the stator currents can be calculated as

$$i_{d1} = \frac{E'_q - U_1 \cos \delta}{X'_1}$$

$$i_{q1} = \frac{U_1 \sin \delta}{X'_1}$$

Then the equations of active and reactive powers of the DFIG stator:

$$P_1 = U_1 I_1 \cos \phi = \frac{E'_q U_1}{X'_1} \sin \delta$$

$$Q_1 = U_1 I_1 \sin \phi = \frac{E'_q U_1}{X'_1} \cos \delta - \frac{U_1^2}{X'_1}$$

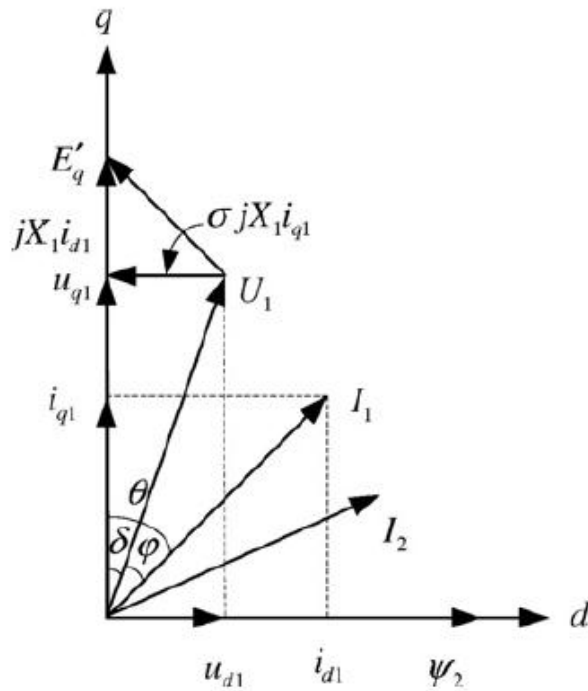


Fig 4.10: Vector diagram of the DFIG

It can be seen that the DFIG has the same expression of active and reactive powers as the synchronous machine. Developing from and adding the stator resistance r_1 , Fig: 4.11 is the single line equivalent circuit of the DFIG. It is similar to that of the synchronous generator except the excitation voltage is different, because it is controlled from a more complex rotor equivalent circuit.

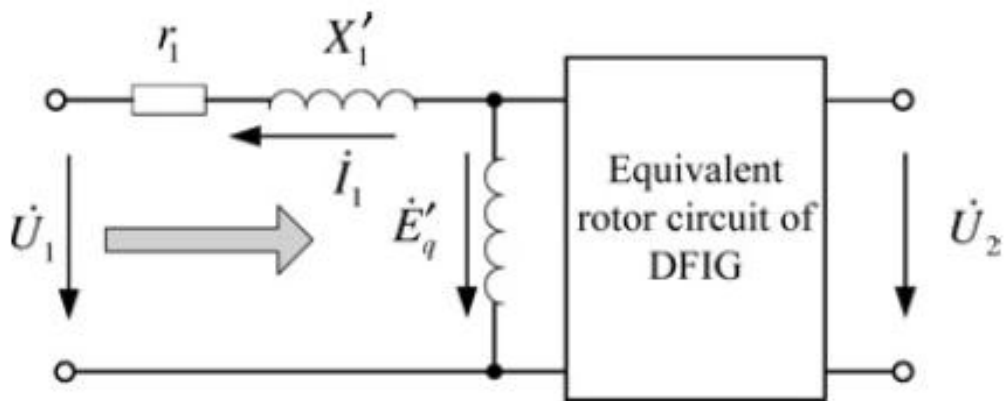


Fig 4.11: Equivalent circuit of DFIG

The power angle in synchronous generator is relatively small in normal operation which is often below 30 degrees. This condition can be also met in DFIG. With this condition the classic synchronous generator theory indicates that the active power transfer depends mainly

on the power angle and the reactive power transfer depends mainly on the voltage magnitude of E'_q , respectively. By similarity of synchronous generator, the control of the stator active power and reactive power of the DFIG can be seen as the control of phase and magnitude of E'_q . The DFIG has a benefit in that the power angle δ (and therefore the active power) is controllable by the rotor converter whereas δ in the synchronous generator is determined by the axis of the field winding.

4.4.2 MODELLING OF THE DFIG ROTOR:

By substituting rotor flux equations into the rotor voltage equations, the rotor voltages can be expressed as:

$$u_{d2} = r_2 \frac{\psi_2}{L_2} + r_2 \frac{L_m}{L_2} + p\psi_2$$

$$u_{q2} = r_2 \frac{L_m}{L_2} i_{q1} + \psi_2 \omega_2$$

Equation above can be re-expressed in the vector form as follows:

$$\mathbf{U}_2 = r_2 \frac{L_m}{L_2} \mathbf{I}_1 + \left(\frac{r_2 + \omega_2 L_2 + p}{\omega_1 L_m} \right) \mathbf{E}'_q$$

Replacing the stator current vector with $\mathbf{I}_1 = (\mathbf{E}'_q - \mathbf{U}_1) / X'_1$, becomes

$$\mathbf{U}_2 = \left(\frac{r_2 L_m}{\sigma L_2 X_1} + \frac{r_2}{X_m} + \frac{\omega_2 L_2}{X_m} + p \right) \mathbf{E}'_q - \frac{r_2 L_m}{\sigma L_2 X_1} \mathbf{U}_1$$

Equation above describes the relationship between the stator voltage vector $\mathbf{U}_1$, the rotor voltage vector $\mathbf{U}_2$ and the internal transient EMF vector $\mathbf{E}'_q$. The stator voltage $\mathbf{U}_1$ is the same as the grid voltage and thus $\mathbf{E}'_q$ can be controlled by $\mathbf{U}_2$.

Unlike the exciter of the synchronous generator which can only adjust the magnitude of the exciter voltage only, the rotor controller of the DFIG can manipulate both the magnitude and the phase angle of $\mathbf{U}_2$ vector. Thus, the active and reactive powers of the DFIG can be controlled by $\mathbf{U}_2$ vector.

MODELLING THE DFIG-BASED WIND TURBINE:

The active power of the DFIG rotor can be expressed as:

$$P_2 = u_{d2}i_{d2} + u_{q2}i_{q2}$$

From the above equations, the active power of the rotor can be expressed as:

$$P_2 = P_{r2} + \frac{\omega_2}{\omega_1} P_1$$

Where,

$$P_{r2} = r_2 i_{d2}^2 + r_2 i_{q2}^2$$

is the power losses associated with the rotor resistance, which is small enough to be ignored. It can be shown that the active power of the rotor depends on the rotor current frequency, stator frequency and the active power of the stator. Depending on the rotor speed ω_r , the rotor current frequency, $\omega_2 = \omega_1 - \omega_r$, can be positive and negative and therefore the rotor power changes direction. The active power of the rotor is positive when the DFIG operates at the sub-synchronous mode ($\omega_1 > \omega_r$) and negative when the DFIG operates at the super-synchronous mode ($\omega_1 < \omega_r$). The grid-side converter, in maintaining the DC-link voltage regulated, feeds or absorbs the slip dependent rotor active power. The reactive power of the grid side converter is set to zero to give a unity displacement factor.

MECHANICAL EQUATION OF MOTION:

The stator voltage vector $\mathbf{U}_1$ rotates at the speed of ω_1 of the grid frequency. The rotating speed of $\mathbf{E}'_q$ is the algebraic sum two speeds: the rotor speed ω_r and the rotor current angular frequency ω_2 . So the equation of the power angle is:

$$\dot{\delta} = (\omega_r + \omega_2) - \omega_1$$

The equation of motion of the rotor is:

$$J\dot{\omega}_r = T_m - T_{em}$$

Where T_m is the input torque from the wind turbine and T_{em} is the electromagnetic torque of the DFIG.

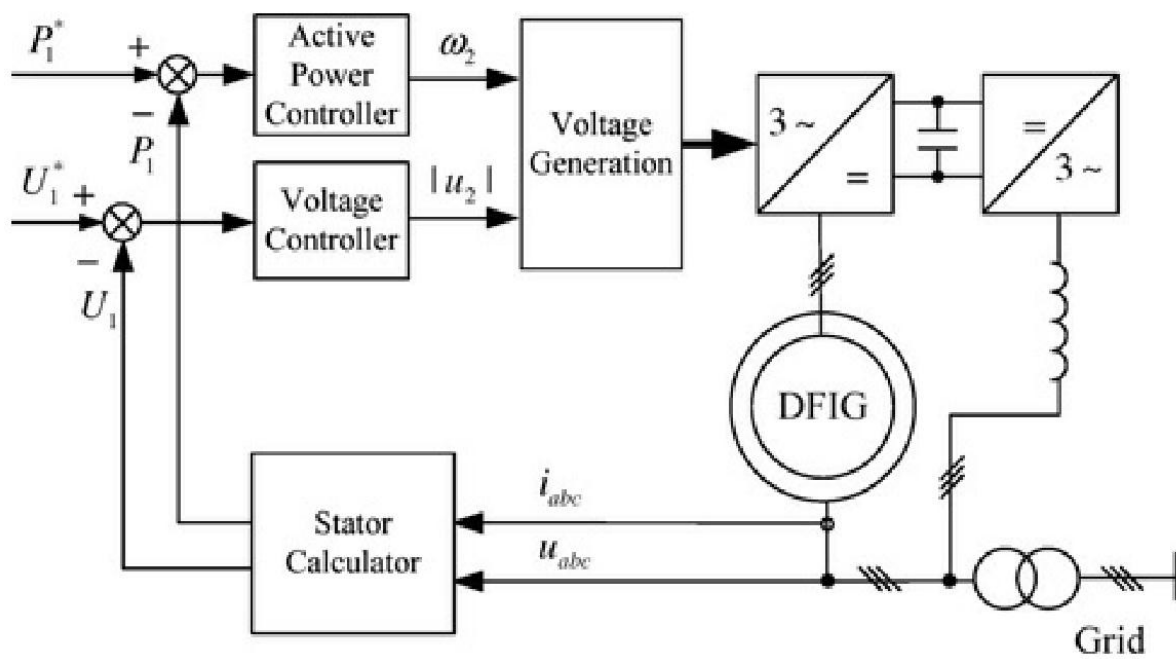


Fig: 4.12 MFC Controller Diagram

5. SIMULATIONS

5.1 PITCH CONTROL ANALYSIS BY MATLAB:

Explanation:

$$P_m = 0.5C_p(\lambda, \beta)\rho AV_w^3$$

Where

P_m is mechanical output power of the wind turbine;

$C_p(\lambda, \beta)$ is the performance coefficient of the turbine;

ρ is the density of air in kg/m^3 ;

A is the swept area of turbine;

V_w is the wind speed (m/s);

λ is the tip speed ratio;

β pitch angle of blade in degrees;

A basic equation used to model $C_p(\lambda, \beta)$:

$$C_p(\lambda, \beta) = c_1 \left(\frac{c_2}{\lambda} - c_3\beta - c_4 \right) e^{\frac{c_5}{\lambda_i + c_6}\lambda}$$

and

$$\frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}$$

The coefficients c_1 to c_6 are: $c_1 = 0.5176$, $c_2 = 116$, $c_3 = 0.4$, $c_4 = 5$, $c_5 = 21$ and $c_6 = 0.0068$. The C_p - λ characteristics, for different values of the pitch angle β , are illustrated below. The maximum value of C_p ($C_{p\max} = 0.48$) is achieved for $\beta = 0$ degree and for $\lambda = 8.1$. This particular value of λ is defined as the Nominal value (λ_{nom}).

PROGRAM:

```

L=0.01:0.1:15;
c1=0.5176;
c2=116;
c3=0.4;
c4=5;
c5=21;
c6=0.0068;
pitch=0:5:25;
for i=1:6
    for p=1:length(L);
        A(p)=1/(L(p)+0.08*pitch(i))-0.035/(pitch(i)^3+1);
        C(p)=c1*(c2*A(p)-c3*pitch(i)-c4)*exp(-c5*A(p))+c6*L(p);
    end
    plot(A(p),C(p));
    hold on;
end
axis([0 15 -0.1 0.5]);
xlabel('\lambda'),ylabel('Cp');

```

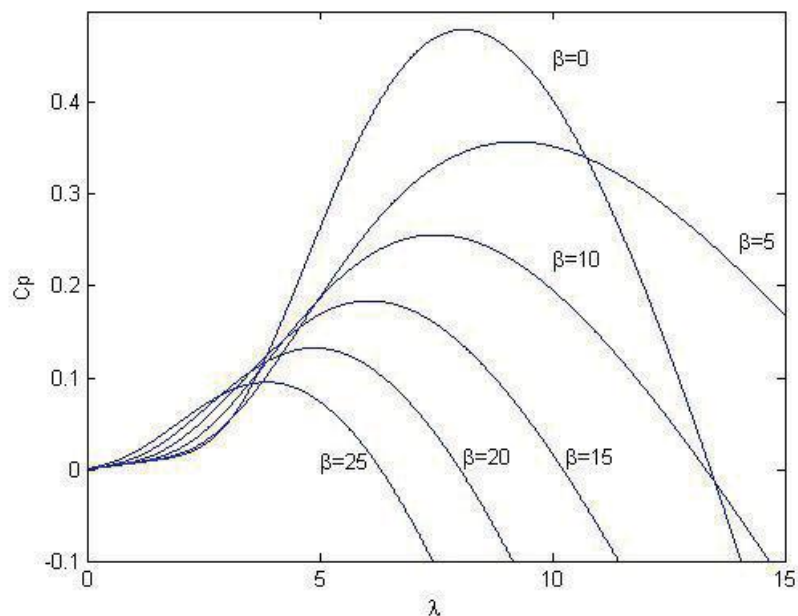


Fig: 5.1 Power coefficients versus tip speed ratio

5.2 MECHANICAL CHARACTERISTICS ANALYSIS BY MATLAB

PROGRAM:

```
% mechanical_characteristics.m
% numerical simulations of the power coefficient of the wind turbine as a function of
the tip
% speed rate and the pitch angle.
c1=0.5176; c2=116; c3=0.4; c4=5; c5=21; c6=0.0068; r0=1.29;
D=40; A=pi*D^2/4;
L=0.01:0.1:15;
b=0;
V=[8,10,12,14,16,18,20];
for k=1:length(V)
    for p=1:length(L);
        AI(p)=1/(L(p))-0.035;
        CP(p)=c1*(c2*AI(p)-c4)*exp(-c5*AI(p))+c6*L(p);
        P(k,p)=(V(k)^3)*CP(p)*r0*A/2;
        n(k,p)=(60/(pi*D))*AI(p)*V(k);
    end;
    hold on;
end;
M=max(P(6,:)); m=max(M); P=P/m; n1=length(L); n2=length(V);
for j=1:n1;
    P(:,n1-j+1)=P(:,j);
end;
PR=P;
for q=1:n2;
    plot(n(q,:), PR(q,:)); hold on
end;
grid; axis([0.1,1.45,-0.1,1.4]);
xlabel('rotational speed(relative units)'),ylabel('power
(relative units)');
```

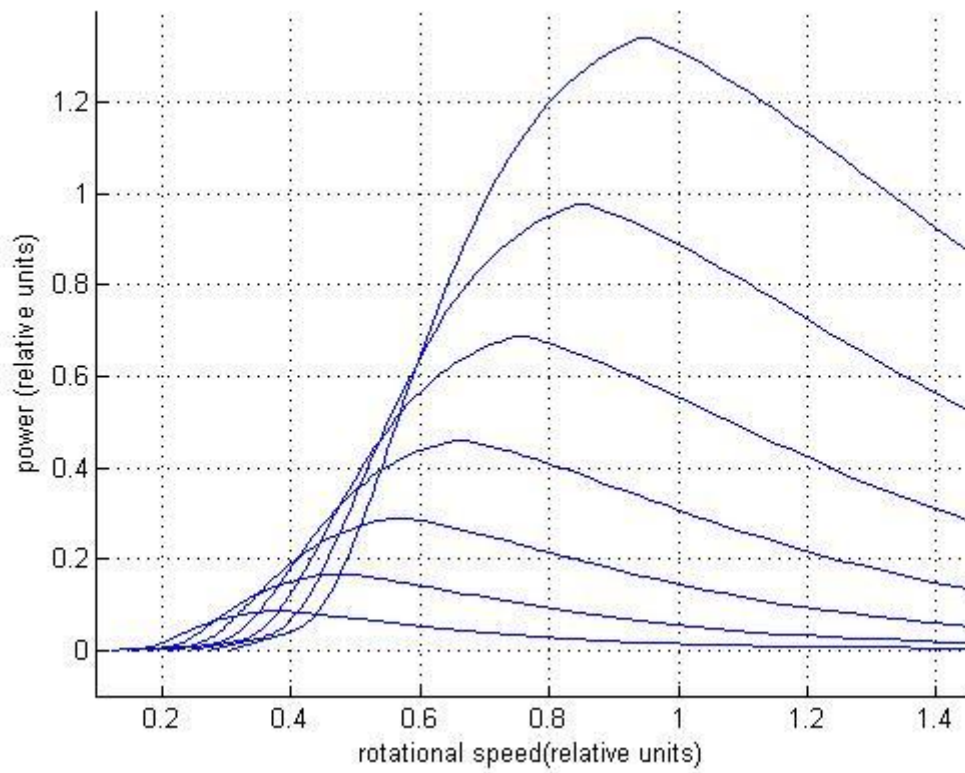
RESULT:

Fig: 5.2 Wind turbine output power vs. rotational speed, with wind speed as parameter

5.3 TORQUE-SLIP CHARACTERISTICS

Torque-slip characteristic when the angle of V_r is 0. $|V_r|$ is changing from -0.05 to +0.05 pu:

MATLAB code:

```
Xls=0.0135;
Xlr=0.0075;
rs=0.00059;
rr=0.00339;
Vs=0.5;
Vr=-0.05:0.01:0.05;
for i=1:11
    s=-1:0.01:1;
    for j=1:201
        T(i,j)=(s(j)*Vs^2-
Vs*Vr(i))*(s(j)*rs+rr)/((s(j)*rs+rr)^2+s(j)^2*(Xls+Xlr)^2);
    end;
    plot(s,T);
end;
axis([-1,1,-15,15])
xlabel('slip'),ylabel('Torque (pu)');
```

RESULT:

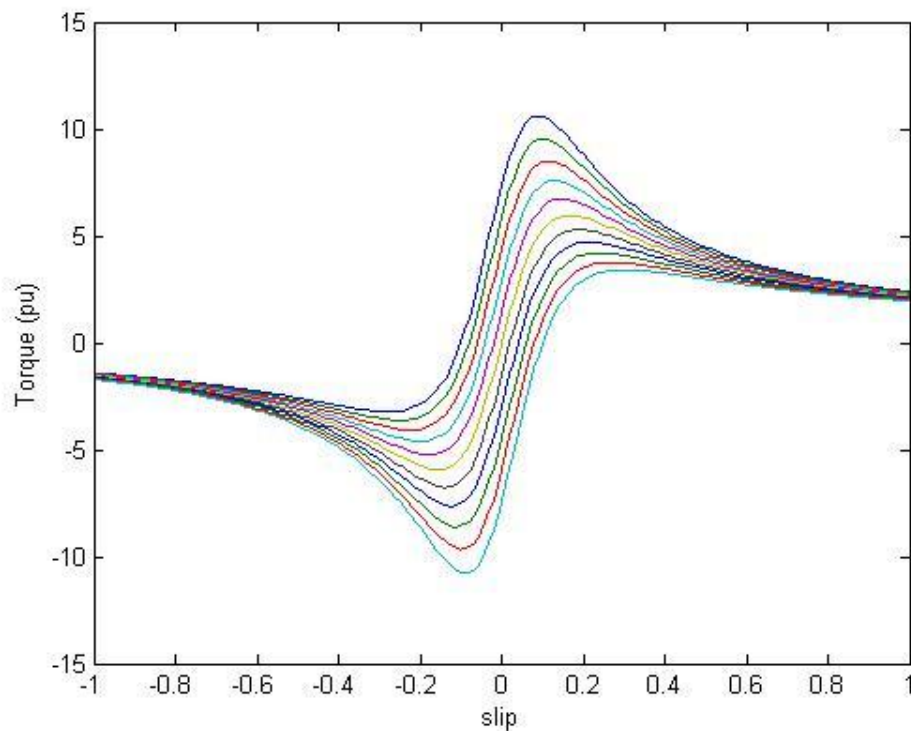


Fig: 5.3 Torque-slip characteristic when the angle of V_r is 0. $|V_r|$ is changing from -0.05 to +0.05 pu.

TORQUE-SLIP CHARACTERISTIC WHEN $|V_r|$ IS 0.05 pu. THE ANGLE OF V_r IS CHANGING FROM -90° TO $+90^\circ$

MATLAB code:

```

clc;
clear all;
Xls=0.0135;
Xlr=0.0075;
rs=0.00059;
rr=0.00339;
Vs=0.5;
Vr=0.05;
angle_deg=-90:20:90;
angle_rad=deg2rad(angle_deg);
for i=1:length(angle_deg)
    s=-1:0.01:1;
    for j=1:201
        T(i,j)=(((s(j)*Vs^2-
Vs*Vr*cos(angle_rad(i)))*(s(j)*rs+rr))+(Vs*Vr*s(j)*(Xls+Xlr)*s
in(angle_rad(i))))/((s(j)*rs+rr)^2+s(j)^2*(Xls+Xlr)^2);
    end;
    plot(s,T);
end;
% axis([-1,1,-15,15])
xlabel('slip'),ylabel('Torque(pu)');

```

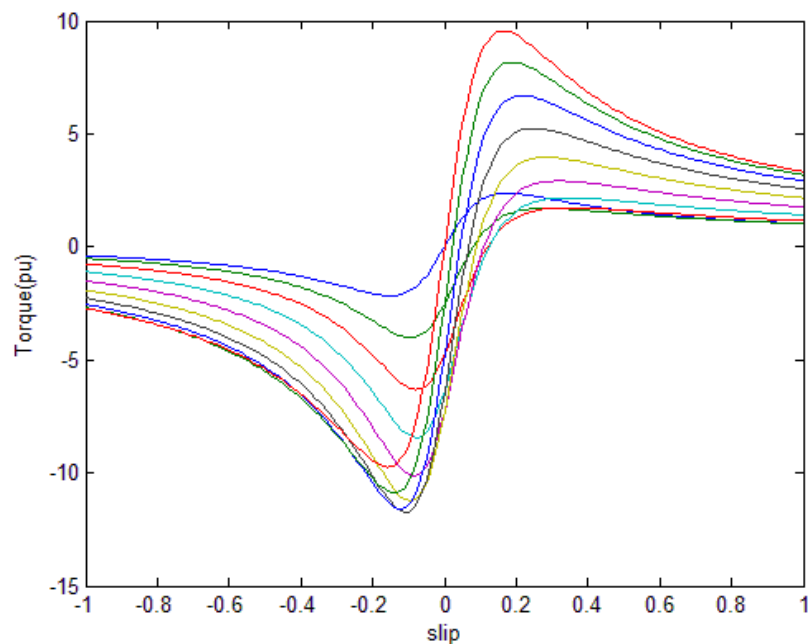


Fig: 5.4 Torque-slip characteristic when $|V_r|$ is 0.05 pu. The angle of V_r is changing from -90° to $+90^\circ$